Easy (10 MCQs)

1.

What is the maximum length for a standard 100BASE-TX Ethernet cable segment?
A) 10 meters
B) 100 meters
C) 1000 meters
D) 10 kilometers
Answer: B
Explanation: 100BASE-TX uses Cat5 cables with a max length of 100 meters.

2.

Which device forwards Ethernet frames based on MAC addresses?
A) Hub
B) Switch
C) Router
D) Repeater
Answer: B
Explanation: Switches use MAC addresses to forward frames only to the intended recipient port.

3.

What does the acronym CSMA/CD stand for?
A) Carrier Sense Multiple Access with Collision Detection
B) Carrier Send Multiple Access with Collision Detection
C) Carrier Sense Multiple Access with Collision Delay
D) Carrier Send Multiple Access with Collision Delay
Answer: A
Explanation: CSMA/CD is the Ethernet access control protocol detecting collisions.

4.

Which wireless standard operates primarily in the 2.4 GHz band and supports up to 11 Mbps?
A) 802.11a
B) 802.11b
C) 802.11g
D) 802.11n
Answer: B
Explanation: 802.11b works at 2.4 GHz and max 11 Mbps.

5.

What is the typical size of a MAC address?
A) 16 bits
B) 32 bits
C) 48 bits
D) 64 bits
Answer: C
Explanation: MAC addresses are 48 bits long.

6.

Which Ethernet device broadcasts all incoming packets to every port?
A) Switch
B) Router
C) Hub
D) Bridge
Answer: C
Explanation: Hubs broadcast all traffic, causing collisions.

7.

Which wireless security protocol is considered the strongest and most secure?
A) WEP

B) WPA
C) WPA2
D) WPA3
Answer: D
Explanation: WPA3 is the latest and most secure protocol.

8.

What does an access point (AP) do in a wireless network?
A) Routes traffic between networks
B) Connects wireless clients to a wired network
C) Amplifies wired signals
D) Assigns IP addresses
Answer: B
Explanation: APs allow wireless devices to connect to wired LANs.

9.

In Ethernet, what is the minimum payload size in a frame?
A) 46 bytes
B) 64 bytes
C) 1500 bytes
D) 1518 bytes
Answer: A
Explanation: Ethernet frames require at least 46 bytes of data.

10.

Which frequency bands are commonly used in Wi-Fi networks?
A) 1.8 GHz and 2.4 GHz
B) 2.4 GHz and 5 GHz
C) 5 GHz and 10 GHz
D) 900 MHz and 1.8 GHz

Answer: B
Explanation: Wi-Fi commonly uses 2.4 GHz and 5 GHz bands.

Medium (30 MCQs)

11.

What is the purpose of the Start Frame Delimiter (SFD) in an Ethernet frame?
A) To indicate the destination address
B) To mark the start of the frame data
C) To check for errors
D) To indicate the source address
Answer: B
Explanation: The SFD indicates the start of the frame's payload.

12.

Which of the following cables is NOT typically used for Ethernet networking?
A) Cat5e
B) Cat6
C) Coaxial cable
D) Fiber optic cable
Answer: C
Explanation: Coaxial cables were used in older Ethernet standards but are obsolete now.

13.

In a wireless network, what is the role of RTS/CTS?

A) Encryption method
B) Collision avoidance technique
C) Frequency hopping algorithm
D) Authentication protocol
Answer: B
Explanation: RTS/CTS reduces collisions by reserving the medium before transmission.

14.

Which of these wireless network modes allows devices to connect without an access point?
A) Infrastructure mode
B) Ad hoc mode
C) Bridge mode
D) Mesh mode
Answer: B
Explanation: Ad hoc mode enables direct peer-to-peer communication.

15.

How does CSMA/CD react when it detects a collision on an Ethernet network?
A) Immediately retries sending the data
B) Stops transmitting and waits a random backoff time
C) Switches to half duplex
D) Switches to full duplex
Answer: B
Explanation: After collision detection, the device waits a random time before retrying.

16.

Which IEEE standard defines VLAN tagging?
A) 802.1D
B) 802.1Q
C) 802.3

D) 802.11ac
Answer: B
Explanation: 802.1Q standardizes VLAN tagging in Ethernet frames.

17.

What is the maximum theoretical throughput of 802.11n Wi-Fi?
A) 54 Mbps
B) 600 Mbps
C) 1 Gbps
D) 11 Mbps
Answer: B
Explanation: 802.11n supports up to 600 Mbps using MIMO.

18.

Which Ethernet feature allows multiple devices to communicate simultaneously without collisions?
A) Half duplex
B) CSMA/CD
C) Full duplex
D) Token passing
Answer: C
Explanation: Full duplex allows simultaneous send and receive, eliminating collisions.

19.

What is the primary function of the Frame Check Sequence (FCS) in an Ethernet frame?
A) Synchronize the clock
B) Identify the source MAC address
C) Detect errors in the frame
D) Indicate the type of payload
Answer: C
Explanation: FCS is a CRC used to detect transmission errors.

20.

What type of antenna is commonly used to provide omnidirectional wireless coverage?
A) Yagi antenna
B) Dipole antenna
C) Parabolic antenna
D) Directional antenna
Answer: B
Explanation: Dipole antennas radiate signals in all directions.

21.

Which device separates collision domains but not broadcast domains?
A) Router
B) Switch
C) Hub
D) Bridge
Answer: D
Explanation: Bridges separate collision domains, but broadcast traffic passes through.

22.

Which wireless encryption protocol replaced WEP due to its weaknesses?
A) WPA2
B) WPA
C) WPA3
D) TKIP
Answer: B
Explanation: WPA was introduced to fix WEP's flaws.

23.

Which of the following is NOT a benefit of using VLANs in an Ethernet network?

A) Reduced broadcast domains
B) Improved security
C) Physical separation of networks
D) Logical segmentation
Answer: C
Explanation: VLANs logically segment a network without physical separation.

24.

Which modulation technique is used in 802.11a and 802.11g Wi-Fi standards?
A) DSSS (Direct Sequence Spread Spectrum)
B) OFDM (Orthogonal Frequency Division Multiplexing)
C) FHSS (Frequency Hopping Spread Spectrum)
D) ASK (Amplitude Shift Keying)
Answer: B
Explanation: OFDM is used for higher data rates in these standards.

25.

What does the Ethernet term "backoff algorithm" refer to?
A) How devices handle collisions and decide when to retransmit
B) Encryption method for Ethernet frames
C) IP addressing method
D) Switch forwarding process
Answer: A
Explanation: Backoff algorithms prevent repeated collisions by random wait times.

26.

What is the standard length of an Ethernet preamble?
A) 8 bytes
B) 7 bytes
C) 6 bytes

D) 5 bytes
Answer: B
Explanation: The Ethernet preamble is 7 bytes of alternating bits.

27.

Which wireless topology involves nodes forwarding data for the network and dynamically self-configuring?
A) Star
B) Mesh
C) Bus
D) Ring
Answer: B
Explanation: Mesh networks provide dynamic routing and self-healing capabilities.

28.

Which device in an Ethernet network operates at Layer 2 and uses MAC addresses?
A) Router
B) Switch
C) Firewall
D) Hub
Answer: B
Explanation: Switches operate at Layer 2 and forward frames based on MAC addresses.

29.

What is the primary reason full-duplex Ethernet does not require CSMA/CD?
A) It uses wireless media
B) There are no collisions possible in full duplex
C) It uses token passing
D) It always sends at half speed
Answer: B
Explanation: Full-duplex links allow simultaneous send and receive, so collisions can't occur.

30.

Which of the following is NOT a wireless frequency band used for Wi-Fi?
A) 900 MHz
B) 2.4 GHz
C) 5 GHz
D) 60 GHz
Answer: A
Explanation: 900 MHz is used by some wireless devices but not common for Wi-Fi.

Hard (10 MCQs)

31.

Which Ethernet standard introduced support for 10 Gbps over copper cabling?
A) 10GBASE-SR
B) 10GBASE-T
C) 10GBASE-LX4
D) 10GBASE-CX4
Answer: B
Explanation: 10GBASE-T allows 10 Gbps over twisted-pair copper (Cat6a or higher).

32.

How does 802.11ax (Wi-Fi 6) improve efficiency in dense environments?
A) By using OFDM only
B) By employing MU-MIMO and OFDMA
C) By increasing transmission power

D) By reducing bandwidth
Answer: B
Explanation: Wi-Fi 6 uses MU-MIMO and OFDMA for better simultaneous multi-user support.

33.

Which layer in the OSI model is responsible for logical link control (LLC) in Ethernet?
A) Physical Layer
B) Data Link Layer
C) Network Layer
D) Transport Layer
Answer: B
Explanation: The Data Link Layer consists of LLC and MAC sublayers.

34.

What is the effect of enabling VLAN tagging (802.1Q) on the Ethernet frame size?
A) Adds 4 bytes to the frame
B) Reduces payload by 4 bytes
C) No effect
D) Adds 8 bytes to the frame
Answer: A
Explanation: VLAN tags add a 4-byte tag field inside the Ethernet frame.

35.

Which of the following is NOT true about CSMA/CA used in wireless networks?
A) It avoids collisions by waiting for acknowledgments
B) It uses RTS/CTS to reduce collisions
C) It detects collisions during transmission like CSMA/CD
D) It includes a random backoff timer
Answer: C
Explanation: CSMA/CA avoids collisions but cannot detect collisions during transmission.

36.

In 802.11 wireless, what does the Network Allocation Vector (NAV) indicate?
A) Frequency channel being used
B) Duration for which the medium is reserved
C) Signal strength of the AP
D) MAC address of the client
Answer: B
Explanation: NAV is used by devices to defer transmissions for a specified duration.

37.

What is the major drawback of hubs compared to switches in Ethernet?
A) Higher cost
B) Limited to half duplex and prone to collisions
C) Complex configuration required
D) Only support fiber optic cables
Answer: B
Explanation: Hubs cause collisions and operate only in half duplex.

38.

Which wireless technology allows frequency hopping to avoid interference?
A) OFDM
B) FHSS
C) DSSS
D) MIMO
Answer: B
Explanation: Frequency Hopping Spread Spectrum changes frequencies rapidly to avoid interference.

39.

What is the minimum frame size of an Ethernet frame and why is it necessary?

A) 64 bytes, to ensure collision detection works properly
B) 46 bytes, to accommodate minimum payload
C) 128 bytes, for addressing purposes
D) 1500 bytes, to support max payload
Answer: A
Explanation: Minimum 64 bytes ensures collisions are detected before transmission completes.

40.

Which of the following wireless security features is unique to WPA3 compared to WPA2?
A) AES encryption
B) Simultaneous Authentication of Equals (SAE)
C) TKIP protocol
D) WEP fallback support
Answer: B
Explanation: SAE provides stronger key exchange and protection against offline dictionary attacks.